Lecture 2
Cyber Security

Dr. Muhammad Anwar

What is Cyber Security

- Information security consists of the policies and practices adopted to prevent unauthorized access, misuse, modification, malfunction, and destruction of data and information.
- Information security involves the authorization of access to data in a network, which is controlled by the network administrator. Each user is assigned an ID and password that allows them access to the network resources within their authority.

Objectives of Information Security

- Confidentiality
- Integrity
- Availability

Objectives of Cyber Security

- **Confidentiality** refers to protecting the information from being accessed by unauthorized persons. In other words, only the people who are authorized to do so can gain access to sensitive information.
- A failure to maintain the confidentiality means that someone who shouldn't have access has managed to get it, through intentional behavior or by accident. Such a failure of confidentiality, commonly known as a **Confidentiality breach**, which normally cannot be remedied. Once the secret has been revealed, there's no way to un-reveal it.

Objectives of Cyber Security

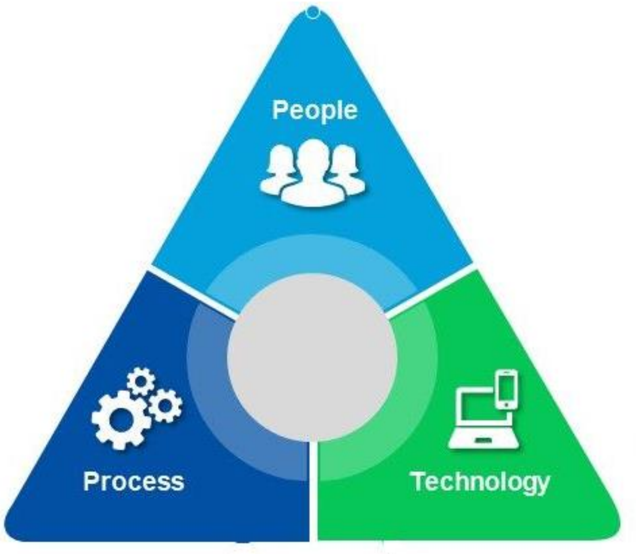
- **Integrity** refers to ensuring the authenticity of information—that the information is not altered, and that the source of the information is genuine.
- Imagine that you have a website and you sell products on that site. Now imagine that an attacker can shop on your web site and maliciously alter the prices of your products, so that they can buy anything for whatever price they choose. That would be a failure of integrity, because your information.

Objectives of Information Security

- **Availability** means that information is accessible by authorized users. If an attackers are not able to access the desired information security, they may try to execute attacks like denial of service that would bring down the server, making the website unavailable to legitimate users due to lack of availability.
- For examples, University website should be available to the students and teachers all the time.
- Availability can be maintained by redundancy.

Foundations of Cyber Security

- People
- Processes
- Technology



Also called People, Process and Technology (PPT) framework.

Foundation of Cyber Security

- **People** are those person who are working on your network that can be network users or network administrators.
- People are the most important part of the network security triangle. Without people, nothing can happen.
- Network administrators should be trained enough to maintained the security in the network.
- Network users should know and follow the security policies of the organization. What is acceptable use of network resources for them? That is, who can access which information? What are their authorities? Are they authorize to copy data on USB drive and take home?

Foundation of Cyebr Security

- A **process** is a series of actions or steps that need to happen in order to achieve a particular goal.
- Processes allow IT professionals to deploy network security systems with optimal efficiency, respond to incidents more quickly, and avoid conflicts that could cause harm to the security system as a whole.
- No matter how many IT professionals are working on maintaining the network security, they will be inefficient if they do not have a process in place to act upon.
- Failure to create and adopt adequate processes may results into disaster on your network security system.
- For example, if someone want to access network from home he/she should use VPN.

Foundation of Information Security

- **Technology** is most important part of network security foundations. Without it, we cannot achieve the optical network security in today's modern world.
- If a network does not have a technology based security system, there would be no way to prevent intrusions from occurring, and one could not possibly maintain a log of individual digital security incidents.

Cyber Security Methods

- **Access control:** You should be able to block unauthorized users and devices from accessing your network. Users that are permitted network access should only be able to work with the limited set of resources for which they've been authorized.
- **Anti-malware:** Viruses, worms, and trojans by definition attempt to spread across a network, and can dormant services on infected machines for couple of days or weeks. The security effort should do its best to prevent initial infection and also root out malware that does make its way onto your network.

Cyber Security Methods

- **Application security:** Insecure applications are often the routes by which attackers get access to your network. You need to employ hardware, software, and security processes to secure such applications.
- **Behavioral analytics:** You should know what normal network behavior looks like so that you can spot anomalies or breaches as they happen.
- **Data loss prevention:** Human beings are inevitably the weakest security link. It is required to implement technologies and processes to ensure that staffers don't deliberately or inadvertently send sensitive data outside the network.

Cyber Security Methods

- **Email security:** Phishing is one of the most common ways attackers gain access to a network. Email security tools can block both incoming attacks and outbound messages with sensitive data.
- **Firewalls:** Most important tool for the network security world, they follow the rules you define to permit or deny traffic at the border between your network and the internet, establishing a barrier between your trusted zone and the wild west outside. They don't exclude the need for a defense-in-depth strategy, but they're still a must-have.

Information Security Methods

- **Intrusion detection and prevention:** These systems scan network traffic to identify and block attacks, often by correlating network activity signatures with databases of known attack techniques.
- **Mobile devices and wireless security:** Wireless devices have all the potential security flaws of any other networked gadget — but also can connect to just about any wireless network anywhere, requiring extra scrutiny.

Cyber Security Methods

- **Network segmentation:** Software-defined segmentation puts network traffic into different classifications and makes enforcing security policies easier. Such as creating different VLANs for different Departments in the organization.

- **VPN:** A tool typically based on IPsec that authenticates the communication between a device and a secure network, creating a secure, encrypted "tunnel" across the public internet.

Cyber Security Basic Terminologies

- **Threat** is a possible danger that might breach the security system and cause possible harm.
- **Adversary or threat agent** is an actor that imposes the threat on a specific asset of network resources.
- **Risk** is an expectation of loss, damage or destruction of network services as a result of a threat.
- **Attack** can be defined as any method, process, or means used to maliciously attempt to compromise the network security.